Diffusion-Driven Mutual Enhancement of Matching and Fusion for Reference-Based Hyperspectral Image Super-Resolution

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***Abstract*—Fusion of hyperspectral and multispectral images has become a mainstream technique for obtaining high spatial resolution hyperspectral images. However, in practical applications, it is challenging to obtain spatially well-matched pairs of low-resolution hyperspectral images (LRHSIs) and high-resolution multispectral images (HRMSIs). To tackle this challenge, we propose a matching-fusion mutual enhancement diffusion network (MFME-DiffNet), which leverages the unpaired multiscale matched images to facilitate the generation of high-quality high-resolution hyperspectral image (HRHSI) through an iterative matching-fusion mechanism. Specifically, MFME-DiffNet comprises two core modules: a gradient-aligned multiscale matching module (GAMM) and a similarity-weighted attention fusion (SWAF) module. GAMM is designed to generate multiscale guidance images using dynamic sliding windows and to perform gradient-based feature alignment, resulting in high-quality auxiliary images that are consistent with the target HSI. SWAF is proposed to adaptively integrate multiscale matched images based on their texture similarity with the HSI through a similarity-weighted fusion strategy, and reweight the fused features using the overall similarity of the matched images, thereby significantly enhancing the texture details of the HSI. GAMM and SWAF are iteratively executed during the reverse process of the diffusion model, forming a dynamic feedback mechanism for matching and fusion. Extensive experimental results demonstrate that the proposed MFME-DiffNet can effectively achieve the HSI–MSI fusion with incomplete spatial correspondence, and is significantly superior to existing advanced methods in terms of visual effects and quantitative evaluation indicators. The code is available at https://github.com/Jiahuiqu/MFME-DiffNet**

***Index Terms*—Diffusion model, hyperspectral image (HSI), multiscale matched images, reference-based super-resolution (Ref-SR).**

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I. Introduction

HYPERSPECTRAL imaging (HSI) acquires data in the form of a 3-D cube, where each pixel contains a high-dimensional spectral signature spanning hundreds of contiguous spectral bands [1], [2]. This rich spectral information has facilitated a broad spectrum of applications, including change detection, target detection, and classification [3], [4]. Despite its advantages, HSI systems inherently suffer from a trade-off between spectral and spatial resolution [5]. Specifically, high spectral resolution is typically achieved at the expense of spatial detail, which can adversely impact the performance of subsequent vision tasks, particularly those requiring fine-grained spatial cues.

To enhance the spatial resolution of HSI, early studies proposed single-image super-resolution (SISR) approaches, which reconstruct HR outputs based solely on LRHSIs [6], [7], [8]. These methods typically employ deep neural networks to learn the inverse of the degradation process, aiming to restore spatial details and textures from the limited input. However, due to the inherently insufficient spatial information and lack of contextual cues, SISR methods often suffer from blurring and structural artifacts, particularly in fine-detail regions, thus limiting their effectiveness in practical scenarios. To address this limitation, fusion-based super-resolution (FISR) methods were subsequently introduced [9], [10]. These approaches rely on paired HR auxiliary images, typically captured by panchromatic or multispectral sensors that are spatially and temporally co-registered with the target HSI, to guide the reconstruction process. Leveraging such strictly paired data enables FISR methods to achieve significant improvements in spatial fidelity. The core idea is to fuse the complementary spatial information from the auxiliary image with the spectral richness of the hyperspectral data, under the assumption of precise pixel-wise alignment between the two modalities. However, this strong dependence on accurately aligned and scene-consistent auxiliary images limits the applicability of FISR approaches in complex or dynamic real-world environments where ideal pairings are rarely available.

Reference-based super-resolution (Ref-SR) has recently emerged. Existing Ref-SR methods have been primarily developed for natural images, which typically employ a single auxiliary image acquired at a different time but from the same spatial location as guidance for image super-resolution [11],

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[12], [13], [14], [15]. However, since HSIs typically cover a large spatial extent with complex and diverse land-cover types, a single matched image is inadequate for providing sufficient spatial details to achieve high-quality reconstruction. To address the aforementioned limitations and further improve reconstruction quality, we utilize multiple matched images with diverse semantic levels as guidance inputs, rather than relying on a single matched image. Furthermore, we aim to incorporate an iterative refinement mechanism in which feedback from preliminary fusion results can be used to progressively optimize the image matching process to improve the overall quality of the reconstructed HSI.

Building upon the aforementioned motivation, we propose a matching-fusion mutual enhancement diffusion network (MFME-DiffNet) for reference-based HSI-SR. MFME-DiffNet employs an iterative matching-fusion mechanism during the inverse diffusion process to effectively utilize unpaired multiscale matched images for high-quality HRHSI generation. Specifically, MFME-DiffNet consists of a gradient-aligned multiscale matching module (GAMM) and a similarity-weighted attention fusion (SWAF) module. GAMM designs a multiscale image matching strategy based on robust image features, where gradient features are used to achieve feature alignment across different scales of matched images, thereby generating high-quality guidance inputs for fusion. SWAF introduces a similarity-weighted fusion strategy that adaptively integrates multiscale matched images based on their texture similarity with the HSI, enhancing fusion in semantically aligned regions. It further reweights fused features using overall similarity to suppress interference from dissimilar guidance, thereby improving hyperspectral image fusion quality. These two modules are jointly optimized through alternating iterations in the reverse diffusion process. This closed-loop iterative mechanism enables mutual enhancement between matching and fusion, progressively enhancing the texture details of HSI. The main contributions are summarized as follows.

1) We propose MFME-DiffNet, which fully leverages the semantic and structural information from unpaired multiscale HR multispectral image (HRMSI) to enhance the spatial resolution of HSI, and performs image reconstruction through a diffusion-model-based iterative matching-fusion mechanism, thereby significantly improving the spatial detail of HSI even in the absence of strictly paired HRMSI.
2) We design a GAMM, which extracts multiscale guidance images using dynamic sliding windows and performs feature alignment based on gradient features to obtain high-quality auxiliary images that are highly consistent with the target HSI.
3) We design an SWAF module, in which a similarity-weighted fusion strategy adaptively integrates multiscale matched images based on their texture similarity with the HSI and reweights the fused features using the overall similarity, thereby effectively enhancing fusion quality.

We arrange the remainder of this article as follows. The related works are presented in Section II. The proposed MFME-DiffNet is introduced in Section III. In Section IV, we present both qualitative and quantitative results obtained from four datasets, along with comprehensive ablation studies. The conclusion is drawn in Section V.

II. Related Work

A. Fusion-Based HSI Super-Resolution

Fusion-based HSI super-resolution methods rely on paired HR auxiliary images captured from the same scene at the same time as the target HSI to guide the reconstruction process. These methods can be divided into component substitution (CS), multiresolution analysis (MRA), Bayesian approaches, matrix factorization, and deep learning-based methods. Among them, deep learning-based methods have attracted increasing attention in recent years [16], [17], [18]. Zhang et al. [19] proposed a dictionary learning (DicL)-based HMF framework that jointly addresses image misregistration and nonoverlapped spectral band reconstruction to mitigate registration errors. Guo et al. [20] proposed an online coupled dictionary learning (OCDL)-based method to generate high-quality HRHSI. Han et al. [21] proposed a dictionary expansion-based fusion method that extends the dictionary learned from overlapping areas to nonoverlapping regions to reconstruct HRHSI. In addition, lots of Transformer and diffusion-based methods have been developed successively [22], [23]. Chen et al. [24] proposed a cyclic transformer that leverages spatial–spectral similarities across modalities to achieve superior performance. Sun et al. [25] introduced MDC-FusFormer, which fuses HSI and MSI features at multiple scales, significantly improving fusion quality. Fu et al. [26] proposed a multiscale spatial–spectral dilated Transformer network to increase the receptive field while reducing computational complexity. Sarmad et al. [27] proposed a diffusion- and fusion-based framework for Sentinel-2 super-resolution, achieving superior spatial consistency and spectral fidelity compared to existing methods. Li et al. [28] proposed a model combining Kolmogorov–Arnold networks (KANs) and diffusion models for image fusion, which effectively captures multimodal relationships and suppresses noise to improve fusion quality. Li et al. [29] proposed a dual conditional diffusion model for hyperspectral–panchromatic image fusion, improving spatial–spectral fidelity and reducing blurring. Dong et al. [30] integrated the physical principles of HSI-SR with DDPM, progressively reconstructing HSI across scales to enhance interpretability and reduce inference time. Wu et al. [31] proposed a registration-fusion binocular diffusion model that progressively reconstructs HRHSI. Zhong et al. [32] employed separate spatial and spectral branches along with an alternating projection fusion module (APFM) and a frequency modulation inter-branch module (FMIM). However, the acquisition of ideal paired images is challenging in real-world scenarios, which, to a certain extent, hinders the practical deployment and broader applicability of FISR methods.

B. Reference-Based Image Super-Resolution

Fusion-based HSI super-resolution methods rely on precise spatiotemporal consistency among multisource images, which imposes significant constraints on data acquisition and limits

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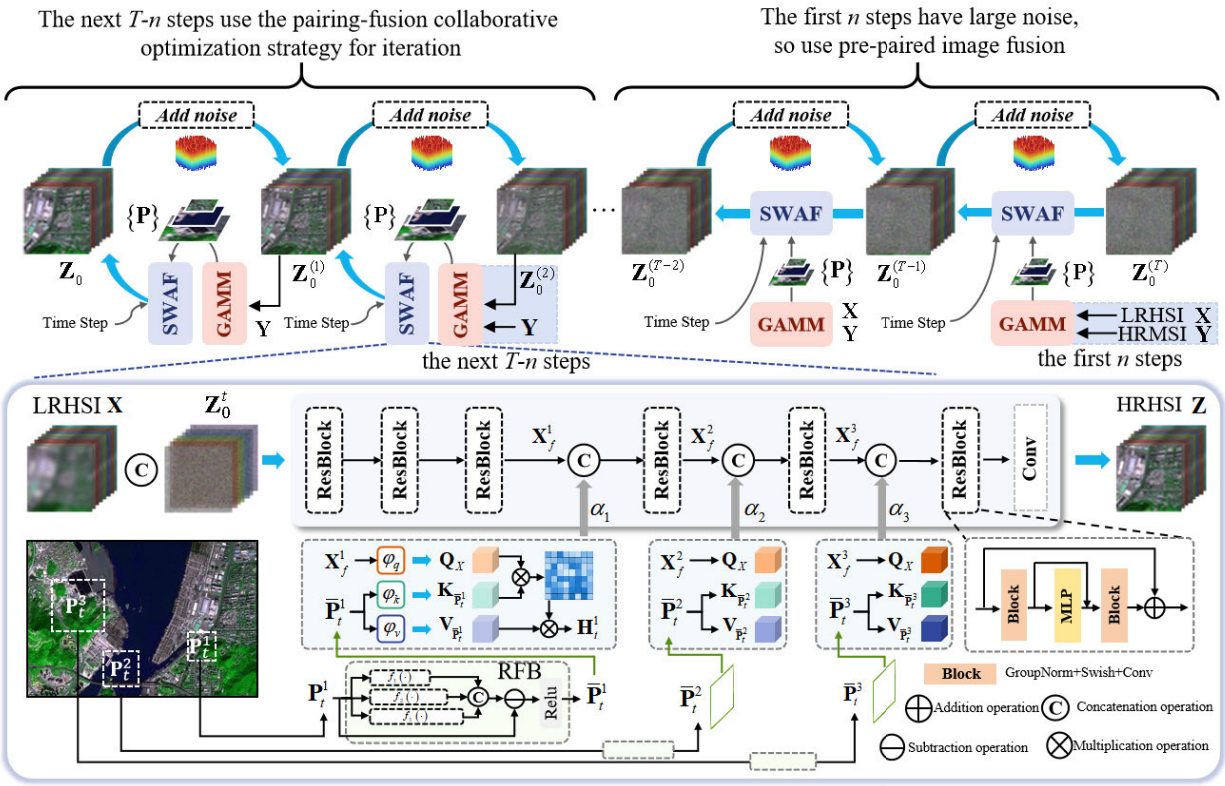


Fig. 1. Illustration of the MFME-DiffNet. At each time step of the reverse diffusion process, the GAMM and the SWAF module are executed alternately. GAMM is designed to generate multiscale matched images, which are subsequently input into SWAF for adaptive fusion of multiscale features based on texture similarity. In the subsequent $T - n$ steps, the output of SWAF is fed into the next GAMM for further refinement. After T iterations of such mutual enhancement, the HR HSI $\mathbf{Z}_0$ is reconstructed.

the applicability of these methods in complex or dynamic real-world scenarios. To enable effective super-resolution in unpaired settings, reference-based image super-resolution leverages an additional unpaired HR reference image to guide the reconstruction process, facilitating the transfer of high-frequency details and texture patterns from the reference to the target image [33], [34]. Currently, several Ref-SR techniques employ the PatchMatch algorithm [35] to establish accurate correspondences between the input and reference images, enhancing the fidelity of detail transfer. For example, Zhang et al. [36] modeled Ref-SR as a neural texture transfer problem and introduced a novel multilevel matching scheme in the feature space, enabling more effective high-frequency detail transfer from reference images while maintaining strong robustness under low-reference similarity. Yang et al. [37] proposed a texture Transformer network that formulates LR and Ref images as queries and keys within a transformer, enabling effective texture transfer and synthesis via hard and soft attention modules. Moreover, several diffusion-based approaches have been developed. Dong et al. [38] proposed a change-aware diffusion framework that leverages land-cover change priors to enhance both content fidelity and reference-guided texture reconstruction. Shi et al. [39] proposed a self-supervised Ref-SR framework that leverages conditional diffusion and SSL representations to generate semantically aligned references, enabling precise texture transfer. Recent works introduced a deformable convolution network (DCN) to align input images and reference images [40], [41], [42]. However, due to the rich and diverse semantic content inherent in HSI, Ref-SR approaches that depend on a single reference image selected via simple matching strategies often fail to provide sufficient guidance for high-quality reconstruction. Therefore, in this work, we employ multiple matched images encompassing various semantic levels to address this limitation.

III. Proposed Approach

As illustrated in Fig. 1, we propose an MFME-DiffNet that utilizes unpaired multiscale matched images to assist the generation of HRHSI via an iterative matching-fusion strategy. MFME-DiffNet comprises of GAMM and SWAF module. GAMM obtains the multiscale matched images through the rotation operator-based gradient features alignment. SWAF generates the fused image by adaptively integrating multiscale matched images with the HSI based on their texture and overall similarity. The two modules are embedded in a diffusion framework for iterative optimization. The reconstruction quality of HRHSI can be significantly improved, even in real-world scenarios where paired auxiliary images are not available.

A. Overall Framework

Given the LRHSI $\mathbf{X} \in \mathbb{R}^{C\times h\times w}$ and the large-width unpaired reference image $\mathbf{Y} \in \mathbb{R}^{c\times \bar{H}\times \bar{W}}$. The aim of Ref-SR is to incorporate the high-frequency details and texture patterns from the HRMSI and preserve the spectral information of

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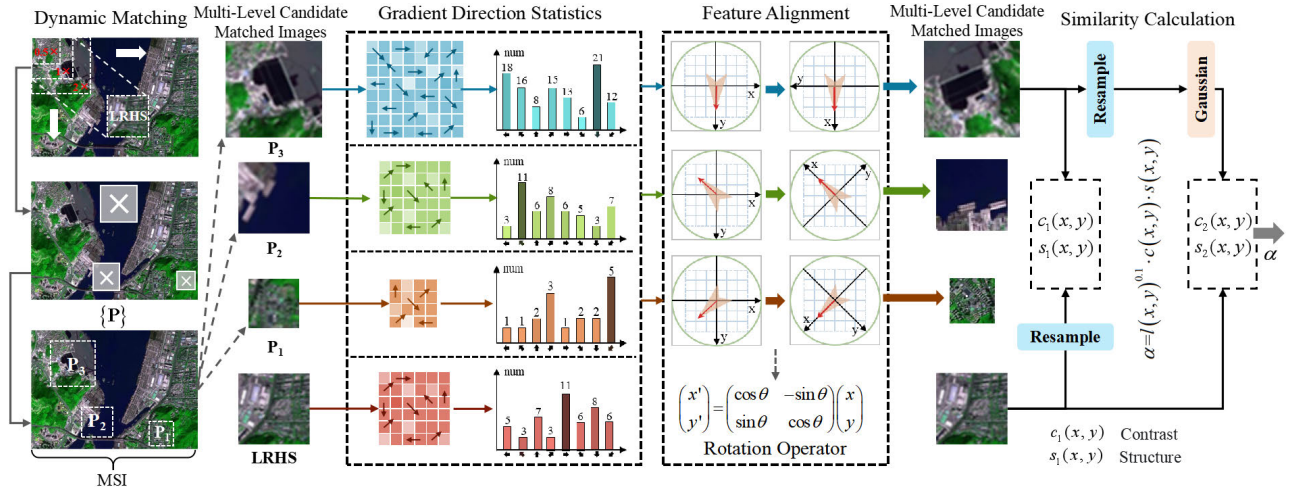


Fig. 2. Framework of GAMM.

the LRHSI to generate an ideal HRHSI $\mathbf{Z} \in \mathbb{R}^{C\times H\times W}$, where $\{c < C\}$, $\{w < W \ll \bar{W}\}$, and $\{h < H \ll \bar{H}\}$ denote the numbers of bands, widths, and heights, respectively. To this end, we propose MFME-DiffNet, which extracts multiple multiscale matched images from the unpaired reference image to assist the generation of high-quality HRHSI through an iterative matching-fusion mechanism. MFME-DiffNet comprises a forward diffusion process that progressive noising the image and a reverse process that performs image matching and fusion interactively.

Specifically, the forward process aims to add Gaussian noise to the HRHSI $\mathbf{Z}_0(\mathbf{Z}_0 = \mathbf{Z})$ step by step, generating a noisy image $\mathbf{Z}_t$. It can be expressed as

$$\mathbf{Z}_t = \sqrt{\overline{\gamma}_t}\mathbf{Z}_0 + \sqrt{1-\overline{\gamma}_t}\boldsymbol{\varepsilon} \tag{1}$$

where $t \in [0, T]$, Gaussian noise $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, hyperparameters $\gamma_t = 1 - \beta_t$, $\overline{\gamma}_t = \prod_{i=0}^{t} \gamma_t$, and $\beta_t \in (0, 1)$ is the variance schedule. After T time steps, this process gradually transforms the data distribution into a standard Gaussian distribution, i.e., $\mathbf{Z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. After generating the noise image $\mathbf{Z}_T$, we design GAMM and SWAF in the reverse reconstruction process. GAMM extracts gradient features for multiscale feature alignment to obtain the best-matched images that are similar to the target image in texture and spatial structure. SWAF leverages the best-matched images obtained from GAMM to assist the improvement of the spatial resolution of LRHSI. These two modules are iteratively optimized to achieve high-quality HRHSI reconstruction during the reverse process.

The reverse process includes a total of T time steps. In the early $T - n$ time steps, the fused images contain considerable noise and lack effective features for accurate image matching. Directly using the fused images for matching would significantly degrade the matching quality, and the original LRHSI performs better for matching at this stage. Thus, we adopt a static matching strategy in the initial reverse stages, where preliminary matched images are obtained using the original LRHSI $\mathbf{X}$ as the source image and HRMSI $\mathbf{Y}$ as the reference image. As the iteration proceeds, the fused images in the later n time steps become less noisy and exhibit clearer spatial structures. Compared with the original LRHSI, which inherently lacks spatial details, the later-stage fused images contain richer spatial information and exhibit higher consistency with the target HRHSI, enabling more precise matching between $\mathbf{Y}$ and the target HRHSI. Therefore, we introduce a dynamic matching update strategy that utilizes the most recent fused image $\mathbf{Z}_0^{(t)}$ at each time step as the source image to perform matching with the reference image $\mathbf{Y}$, enabling the extraction of more spatially aligned matched images relative to the target HRHSI. The matching process during the reverse stage can be represented as

$$\{\mathbf{P}_t\} = \begin{cases} \text{GAMM}(\mathbf{X}, \mathbf{Y}), & t > n \ (\text{Static Matching}) \\ \text{GAMM}\left(\mathbf{Z}_0^{(t)}, \mathbf{Y}\right), & t \leqslant n \ (\text{Dynamic Matching}) \end{cases} \tag{2}$$

where $\{\mathbf{P}_t\}$ is the set of matching images at time step t, and GAMM$(\cdot)$ is the GAMM module for matching process.

Based on the matching results, SWAF then carries out feature fusion and feeds the fused result back to GAMM, thereby optimizing the matching strategy for the next iteration. The result of the current time step can be represented as

$$\mathbf{Z}_0^{(t-1)} = \text{SWAF}\left(\mathbf{Z}_0^{(t)}, \mathbf{X}, \{\mathbf{P}_t\}\right) \tag{3}$$

where SWAF$(\cdot)$ is the SWAF module for fusion operation. The ideal HRHSI $\mathbf{Z}_0$ can be obtained after T iterations.

B. Gradient-Aligned Multiscale Matching Module

As shown in Fig. 2, we design GAMM to achieve structural and textural similarity matching for unpaired LRHSI and HRMSI through gradient feature alignment. Since HSI contains rich and diverse semantic information, a single unpaired matched image cannot effectively compensate for spatial texture details, thereby limiting the potential improvement in fusion performance. To address this limitation, GAMM employs a multiscale dynamic window matching strategy, which dynamically adjusts the sliding window size according to the input LRHSI to effectively mitigate the scale discrepancies between HSI and MSI. This strategy not only avoids the loss of large-scale object details or misalignment of small-scale structures caused by single-scale matching, but

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also enhances matching flexibility and accuracy through the selection of multiscale candidate images.

Specifically, GAMM adopts a sliding window search strategy to determine candidate matched images. To obtain candidates at different scales, the window size is dynamically adjusted according to the size of the target HSI and set to $0.5\times$, $1\times$, and $2\times$ of its dimensions, respectively. The resulting candidates at the three scales are denoted as P_1, P_2, and P_3.

Since the gradient information can effectively characterize the edge and texture structure of an image, it serves as a robust descriptor compared with raw pixel intensities. However, gradient intensity differs between HR and LR images, and directly comparing their gradient intensity can lead to significant errors. In contrast, gradient direction reflects the ratio between horizontal and vertical gradient magnitudes, which can effectively mitigate the impact of variations in gradient intensity across different images. Therefore, we extract cross-spectral structural cues by computing the gradient directions of both HSI and MSI, which enables reliable structural matching and supports the selection of the MSI candidate that is most consistent with the HSI in unpaired and multiscale settings. Specifically, both the target HSI and the candidate MSI are first projected into single-band structural representations using PCA, which preserves the shared spatial structures across spectral channels and provides a consistent 2-D representation for subsequent computation of gradient direction. The gradient direction of the image at the pixel point (x, y) can be expressed as

$$\theta(x,y) = \tan^{-1}\left[\frac{p(x,y+1) - p(x,y-1)}{p(x+1,y) - p(x-1,y)}\right] \tag{4}$$

where $p(x, y+1)$, $p(x, y-1)$, $p(x+1, y)$, and $p(x-1, y)$ are the pixel values at coordinates $(x, y+1)$, $(x, y-1)$, $(x+1, y)$, $(x-1, y)$, respectively, and $\theta(x, y)$ denotes the gradient orientation at pixel (x, y) in the image. The dominant direction of each image is then obtained by evaluating the gradient at all pixel locations and selecting the orientation with the highest occurrence frequency, thereby enabling a statistically reliable basis for subsequent feature alignment and similarity computation. The dominant directions of the HSI and the candidate MSI are, respectively, represented by θ_I and $\theta_{P_i} (i = 1, 2, 3)$.

Considering that the ideal matched MSI should have the same gradient direction as the target HSI, we introduce a rotation operator to re-correct the direction of the candidate MSI. The rotation operator is defined as follows:

$$\mathbf{P}_i' = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \mathbf{P}_i \tag{5}$$

where θ represents the angle difference between θ_{P_i} and θ_I, and $\mathbf{P}_i'$ represents the candidate MSI after rotation.

To quantitatively evaluate the spatial structural similarity (SSIM) between the rotated candidate MSI and the target HSI, we propose an improved SSIM (ISSIM) index for the calculation of GAMM matching scores. ISSIM is optimized based on the traditional SSIM index, which measures similarity based on luminance, contrast, and structural components. However, considering the possible content differences between HRMSI and LRHSI in unpaired scenarios, our goal shifts from enforcing exact spatial–spectral consistency to maximizing the effectiveness of structural information in improving spatial resolution. In this context, luminance information has low discrimination for reliable matches and may even introduce noise interference in similarity calculations. Therefore, ISSIM improves the original SSIM formula by reducing the weight of the luminance term while retaining the emphasis on contrast and structural features. Specifically, we reduce the exponent of the luminance term to 0.1, obtaining the following formula:

$$\mathrm{ISSIM}\left(\mathbf{I}, \mathbf{P}_i'\right) = \left[l\left(\mathbf{I}, \mathbf{P}_i'\right)^{0.1} \cdot c\left(\mathbf{I}, \mathbf{P}_i'\right)^{1} \cdot s\left(\mathbf{I}, \mathbf{P}_i'\right)^{1}\right] \tag{6}$$

where $\mathbf{I}$ represents the target HSI, which can be the input LRHSI $\mathbf{X}$ or the intermediate reconstruction result $\mathbf{Z}_0^{(t)}$ during the reverse diffusion process, $l(\cdot)$ represents the luminance comparison, $c(\cdot)$ denotes the contrast comparison, and $s(\cdot)$ is the structure comparison. This improved similarity metric enables the network to prioritize structural alignment, thereby enhancing spatial details more effectively. During the similarity calculation phase, we first resample the target HSI to the same spatial resolution as the candidate MSI and calculate the ISSIM of the two. Subsequently, we resample the candidate MSI to the spatial resolution of the target HSI and recalculate the ISSIM. Finally, we take the average of the two ISSIM scores to obtain the matching confidence $\alpha_i (i = 1, 2, 3)$. Among all candidate MSIs obtained via the sliding window, we select the one with the highest ISSIM score as the optimal match to guide the subsequent reconstruction process.

C. SWAF Module

After obtaining highly similar unpaired multiscale matched images, how to effectively extract similarity texture features and integrate them into the LRHSI process is the key to improving the fusion performance. To this end, we propose SWAF, which uses a similarity-weighted fusion strategy to integrate multiscale matched images by computing a similarity matrix between the matched images and HSI, effectively enhancing fusion performance in highly similar regions. Moreover, to mitigate the adverse impact of dissimilar images, SWAF uses a similarity coefficient α between the matched images and the target HSI to reweight the fused features, thereby suppressing interference from low-similarity guidance. The fusion process can be described as follows:

$$\mathbf{Z}_0^{(t-1)} = \mathrm{SWAF}\left(\mathbf{Z}_0^{(t)}, \mathbf{X}, \mathbf{P}_t^1, \mathbf{P}_t^2, \mathbf{P}_t^3\right) \tag{7}$$

where $\mathbf{Z}_0^{(t-1)}$ and $\mathbf{Z}_0^{(t)}$ represent the reconstruction results at step $t-1$ and step t, respectively, while $\mathbf{P}_t^i$ denotes the matched images obtained from GAMM at step t.

To avoid the background area in the matched image introducing redundant information during the fusion process and affecting the fusion efficiency, the matched images $\mathbf{P}_t^i (i = 1, 2, 3)$ are not directly fed into the fusion network. Instead, their high-frequency features are first extracted using specific operators. Specifically, a multilevel convolution block is designed to perform blur modeling and low-frequency features extraction. The high-frequency features can be obtained by filtering out the low-frequency components. The extraction

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process of the high-frequency features can be formulated as follows:

$$\begin{aligned}\bar{\mathbf{P}}_t^i &= \mathrm{RFB}\left(\mathbf{P}_t^i\right) \\ &= \mathrm{RFB}\left(\mathbf{P}_t^i - \left[f_1\left(\mathbf{P}_t^i\right), f_2\left(\mathbf{P}_t^i\right), f_3\left(\mathbf{P}_t^i\right)\right]\right)\end{aligned} \tag{8}$$

where $\bar{\mathbf{P}}_t^i (i = 1, 2, 3)$ are the extracted high-frequency features at the tth time step, RFB$(\cdot)$ refers to the receptive field block, $f_1(\cdot)$, $f_2(\cdot)$, and $f_3(\cdot)$ construct the multilevel convolution block, $[\cdot, \cdot]$ represents the concatenation operation, and ReLU is the activation function.

Since $\mathbf{P}_t^i$ is not perfectly aligned with the LRHSI, it may contain redundant or irrelevant spatial information. To ensure that only relevant texture features are injected, we compute the global spatial similarity between the LRHSI and $\mathbf{P}_t^i$, and extract the spatially similar components as the guidance feature, denoted by $\mathbf{H}_t^i$. The computation of $\mathbf{H}_t^i$ is defined as follows:

$$\begin{aligned}\mathbf{H}_t^i &= \frac{\mathbf{Q}_\mathbf{X}\left(\mathbf{K}_{\mathbf{P}_t^i}\right)^T}{\sqrt{B}}\mathbf{V}_{\mathbf{P}_t^i} \\ \mathbf{Q}_\mathbf{X} &= \varphi_q\left(\mathbf{X}_f^i\right) \\ \mathbf{K}_{\mathbf{P}_t^i} &= \varphi_k\left(\mathbf{P}_t^i\right) \\ \mathbf{V}_{\mathbf{P}_t^i} &= \varphi_v\left(\mathbf{P}_t^i\right)\end{aligned} \tag{9}$$

where $\mathbf{X}_f^1$ denotes the features of the LRHSI extracted by three residual blocks (ResBlocks), $\mathbf{X}_f^i (i \geq 2)$ represents the feature obtained from the ResBlock using the previous fused features as input, B is a scaling factor, and $\varphi_q(\cdot)$, $\varphi_k(\cdot)$, and $\varphi_v(\cdot)$ denote convolutional operators used to generate the query, key, and value embeddings, respectively.

In addition, to effectively utilize multiscale matched MSI, we propose a hierarchical progressive weighted fusion strategy. At different scales, the guided feature $\mathbf{H}_t^i$ is adaptively weighted according to the matching confidence $\alpha_i (i = 1, 2, 3)$ obtained by GAMM, thereby achieving more robust and accurate texture detail fusion. The process can be expressed as

$$\begin{aligned}&\mathbf{Z}_0^{(t-1)} \\ &= \mathrm{ResB}^3\left(\left[\mathrm{ResB}^2\left(\left[\mathrm{ResB}^1\left(\left[\mathbf{X}_f^1, \alpha_1\mathbf{H}_t^1\right]\right), \alpha_2\mathbf{H}_t^2\right]\right), \alpha_3\mathbf{H}_t^3\right]\right)\end{aligned} \tag{10}$$

where ResB$(\cdot)$ denotes the ResBlock (ResB), and $[\cdot, \cdot]$ represents channel-wise concatenation. Specifically, taking the first layer of ResB as an example, given the input $\mathbf{X}$ (obtained from $[\mathbf{X}_f^i, \mathbf{H}_t^i]$), the implementation of ResB can be expressed as the following formulations:

$$\begin{aligned}\mathbf{Z} &= \mathrm{ResB}\left(\mathbf{X}\right) \\ &= \mathbf{X} + \Phi_2\left(\Phi_1\left(\mathbf{X}\right) + \mathrm{MLP}\left(\Phi_1\left(\mathbf{X}\right)\right)\right)\end{aligned} \tag{11}$$

where $\mathbf{Z}$ is the output feature of ResB, $\Phi_1(\cdot)$ and $\Phi_2(\cdot)$ represent convolutional blocks composed of group normalization, Swish activation, and convolution, and MLP$(\cdot)$ generates feature-wise affine modulation parameters for conditional feature transformation.

Through multilevel fusion, the features from the matched images are effectively injected, enhancing the texture representation of the HSI and ultimately generating the desired HRHSI.

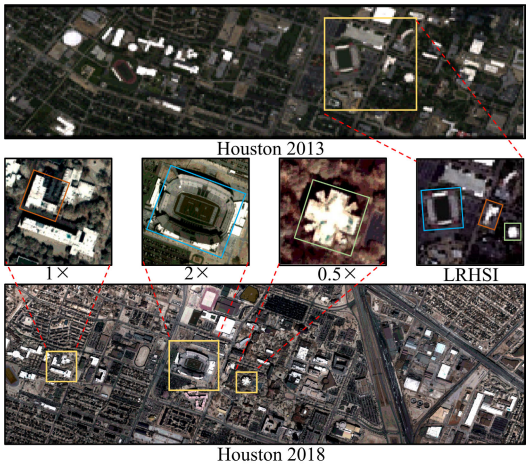


Fig. 3. Houston dataset and multiscale matched images.

D. Training Objective

The proposed MFME-DiffNet aims to reconstruct the target HRHSI through a sequence of registration and fusion steps in an end-to-end learning framework. The overall network is optimized using the $\mathcal{L}_1$ loss function. Accordingly, the consistency constraint used during training to guide the reconstruction of high-quality HRHSI is defined as follows:

$$\mathcal{L}_1 = \frac{1}{N}\sum_{n=1}^{N}\left(\left|\left|\mathbf{Z}^n - \mathbf{Z}_0^n\right|\right|_1\right) \tag{12}$$

where N is the number of training samples, $\mathbf{Z}_0^n$ is the predicted HRHSI with the nth training samples {LRHSI $\mathbf{X}^n$}, and $\mathbf{Z}^n$ is the corresponding GT.

Algorithm 1 Reverse Denoising Process of MFME-DiffNet

Input: Noisy image $\mathbf{Z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, LRHSI $\mathbf{X}$, HRMSI $\mathbf{Y}$, total steps T, threshold n
Output: High-resolution HSI $\mathbf{Z}_0$
Initialize $\mathbf{Z}_T$ from Gaussian noise;
while $t \leftarrow T$ *to* 1 **do**
  if $t > n$ *(high noise stage)* **then**
    Set source image = $\mathbf{X}$ `// Static matching`
  else
    Set source image = $\mathbf{Z}_0^{(t)}$ `// Dynamic matching`
  Extract multiscale candidate matched images from $\mathbf{Y}$ by sliding windows;
  Align gradient orientations of candidate matched images with source image by Eq.(5) ;
  Select the best matched images by ISSIM $\rightarrow \mathbf{P}_t$ by Eq. (6);
  Extract high-frequency features from $\bar{\mathbf{P}}_t \rightarrow \mathbf{P}_t$ by Eq. (8) ;
  Compute attention between the features $\mathbf{X}_f$ and $\bar{\mathbf{P}}_t \rightarrow \mathbf{H}_t$ by Eq. (9);
  Fuse $\mathbf{X}_f$ and $\mathbf{H}_t$ progressively across scales $\rightarrow \mathbf{Z}_0^{(t-1)}$ by Eq. (10);
return $\mathbf{Z}_0$;

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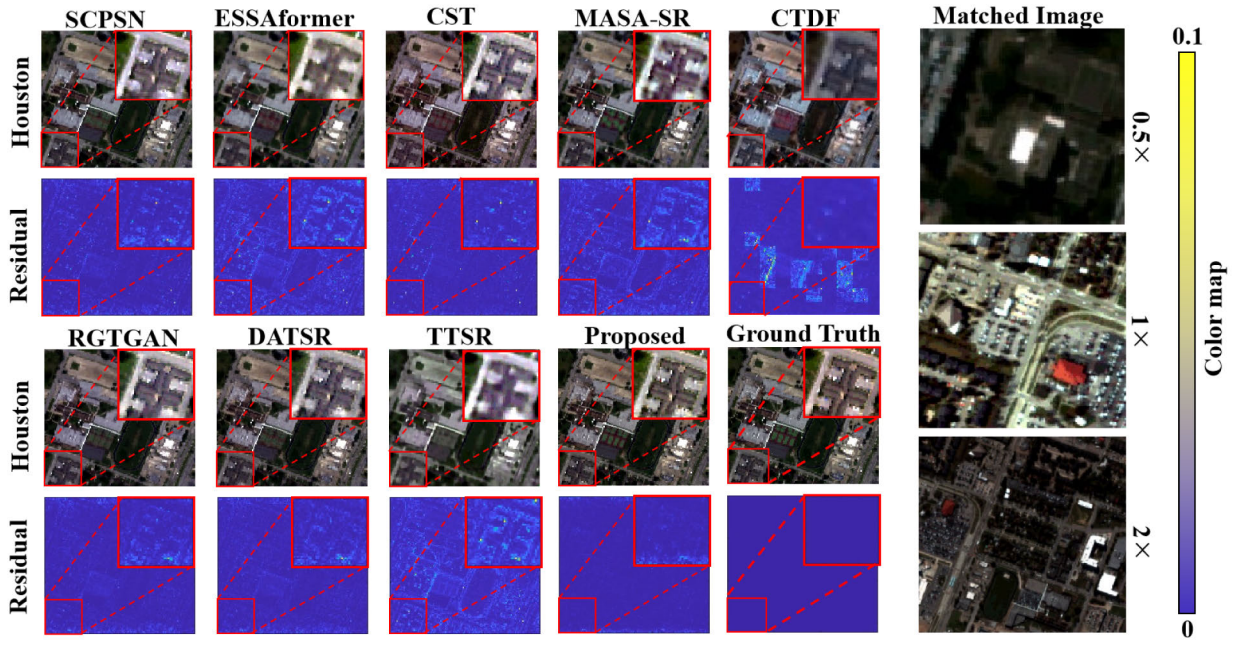


Fig. 4. Visualization results of different methods on Houston dataset with GT.

The algorithmic procedure of the reverse process is shown in Algorithm 1.

IV. Experiment

A. Dataset

To illustrate the effectiveness of the proposed method, we conduct experiments on four datasets, namely Houston, Pavia, Chikusei, and XDU-Liyukou datasets. As shown in Fig. 3, the LRHSI and HRMSI were deliberately chosen from different regions. The HRMSI, LRHSI used in the Houston, Pavia, and Chikusei datasets are generated according to Wald's protocol [43]. Specifically, the simulated LRHSIs are created by applying a 9×9 Gaussian filter and then by downsampling, every four pixels in the spatial dimension of the reference images. The simulated HRMSIs are generated by projecting the reference HSI onto the spectral response functions of the multispectral sensor, thereby producing images with reduced spectral resolution but higher spatial resolution. The XDU-Liyukou dataset is a real-world dataset.

1) Pavia Dataset: The unpaired dataset was constructed based on the Pavia Center and Pavia University scenes. Specifically, HRHSI is derived by partitioning the original Pavia Center dataset into 252 patches, each with a spatial resolution of 160 $\times$ 160 pixels and 102 spectral bands. LRHSIs are obtained by bicubic downsampling each HRHSI patch by a factor of 4. To simulate the unpaired setting and introduce realistic spatial discrepancies, we replace the original MSI of Pavia Center with the HRMSI from the Pavia University dataset. The entire HRMSI with a spatial resolution of 640 $\times$ 320 pixels and 4 spectral bands is used.

2) Houston Dataset: The unpaired dataset is constructed based on the Houston dataset. Fig. 3 shows the Houston dataset and its corresponding multiscale matched images. The HRHSI are extracted from the Houston 2013 dataset by cropping 147 patches, each with a spatial resolution of 160 $\times$ 160 pixels and 144 spectral bands. The corresponding MSI is obtained from the Houston 2018 dataset, which offers updated urban observations and broader spatial coverage. To ensure a more consistent spatial resolution, the MSI is uniformly downsampled by a factor of 2, yielding an image of size 320 $\times$ 1120 $\times$4.

3) Chikusei Dataset: The Chikusei dataset is prepared by selecting a central region of size 2048 $\times$ 2048 $\times$ 128 from the original image, with peripheral edge areas removed to ensure data quality. To establish an unpaired fusion scenario, the selected region is divided into two nonoverlapping halves along the vertical axis. Specifically, the left half of the HSI and the right half of the MSI are discarded, ensuring no direct spatial correspondence between modalities. From the right-side HSI region, a total of 196 patches of size 256 $\times$ 256 $\times$ 128 are extracted as HRHSI samples, and their corresponding LRHSI counterparts are generated via downsampling to 64 $\times$ 64 $\times$ 128.

4) XDU-Liyukou Dataset: [44][1] The XDU-Liyukou dataset, independently collected by our research group, contains both HSI and RGB images. The LRHSI, with a spatial size of 250 $\times$ 125 pixels, was acquired on April 6, 2023, using the Headwall VNIR sensor. It spans a spectral range of 0.4 $\sim$ 1.0 μm, with 194 bands for analysis. The RGB image, with a resolution of 750 $\times$ 375 pixels, was obtained on November 18, 2022, using the FLIR DUO Pro R sensor. The XDU-Liyukou dataset was originally developed to support multimodal change detection research. The two images were acquired at different times, capturing land-cover changes in Liyukou Village, Huyi District, Xi'an, Shaanxi Province. Significant differences in land-cover types between the two images result in the absence of strict spatial or semantic correspondence, making the dataset suitable for

[1]https://github.com/Jiahuiqu/XDU-Liyukou-dataset

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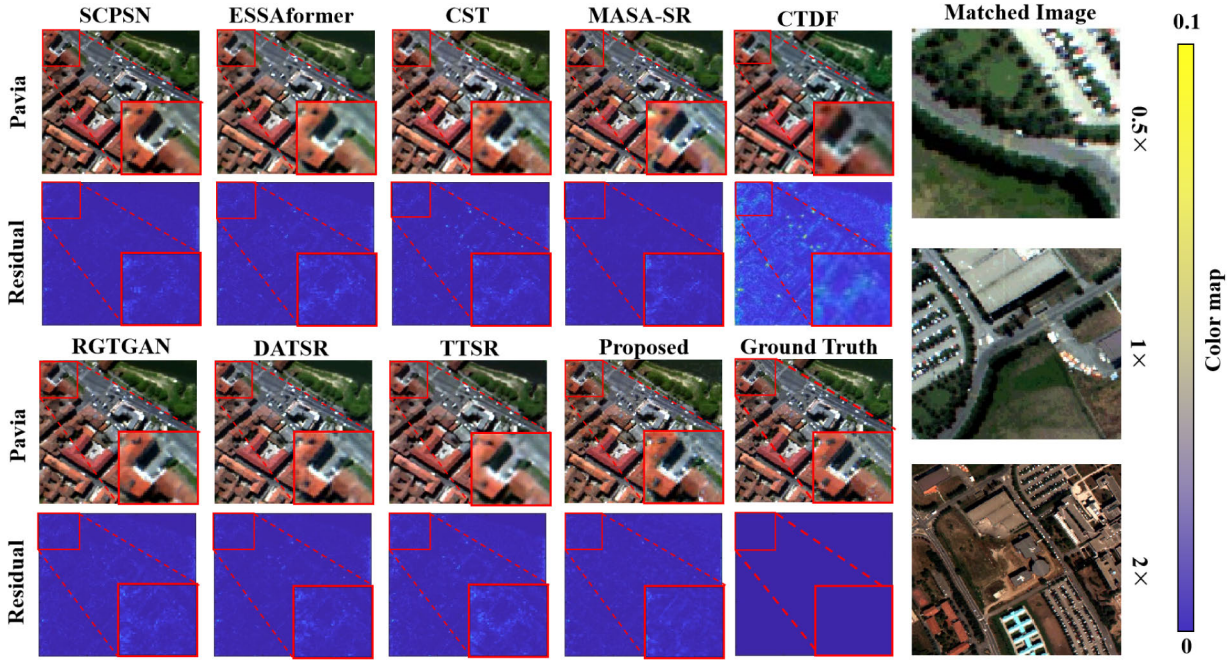


Fig. 5. Visualization results of different methods on Pavia dataset with GT.

both change detection and unpaired multisource image fusion research.

To mitigate the risk of data leakage, taking the Pavia Center dataset as an example, we first partition the image into 12 nonoverlapping large patches of size 160×320, from which nine large patches are randomly selected for training, and the remaining three are retained for testing. Each of the nine large patches selected for training is further subdivided into 21 smaller patches of size 160×160 with a step size of 8, yielding a total of 189 training samples to augment the dataset. Similarly, the three large patches selected for testing are further subdivided to obtain 63 testing samples. This ensures that the training and testing data are strictly independent and nonoverlapping. The same strategy is adopted for the other datasets, thereby eliminating the risk of data leakage.

B. Implementation Details

We conducted the experiments with the PyTorch framework and trained on four NVIDIA GeForce RTX 3090 GPUs. The experiments were conducted with a batch size of 8 and 200k iterations on all datasets. The Adam optimizer was employed for the optimization process, with a maximum learning rate of 0.0001. The time step T was set to 2000, and the hyperparameter sequence $\{\beta_1, \beta_2, \ldots, \beta_n\}$ was defined with uniform growth ranging from 0 to 0.02.

C. Experimental Results

To validate the effectiveness of MFME-Diffnet, we compare it against several existing methods, including SCPSN [45], ESSAformer [46], CST [6], MASA-SR [14], CTDF [47], RGTGAN [48], DATSR [49], and TTSR [37]. Among these, SCPSN, ESSAformer, and CST are SISR-based methods, while the others are Ref-SR-based approaches. Due to the limited research on Ref-SR tasks for HSI, we adopt several methods originally designed for natural images, retaining their core frameworks and making minor modifications to enable processing of HSI.

For the experiments on Pavia, Houston, and Chikusei datasets, the fusion performance was evaluated using five metrics: PSNR, CC, SAM, RMSE, and ERGAS.

For XDU-Liyukou (real-word) dataset without real HR hyperspectral images [ground truth (GT)], we adopt two types of evaluation strategies. First, we use the original LRHSI as GT, spatially downsample the LRHSI and RGB images to construct a test pair with pseudo-GT, and utilize reduced-resolution metrics (PSNR, CC, SAM, RMSE, and ERGAS) for quantitative evaluation. Second, we directly use the original LRHSI and RGB images as model input, and use a full-resolution metric for evaluation without GT. We further add the following explanation to this. Quality with no reference (QNR) is a typical nonreference image quality evaluation index. It measures the deviation between the fusion result and the input image in terms of spectral preservation and spatial detail consistency through spectral distortion and spatial distortion. However, it should be noted that in the unpaired condition of this task, the HRMSI lacks a direct correspondence in the spatial dimension with the output HRHSI, so the terms in QNR used to measure spatial distortion are not applicable. We only use its spectral distortion index D_λ to objectively evaluate the fusion quality in a real scenario. Its value range is 0–1. The closer to 0, the higher the fusion quality.

1) Qantitative and Qualitative Evaluation: The quantitative comparison between the proposed method and existing approaches is reported in Table I. The SISR methods consistently yield inferior performance compared to Ref-SR methods on both the Houston and Pavia datasets. Although Ref-SR approaches achieve certain improvements over SISR baselines, their guidance capability remains limited, as most rely

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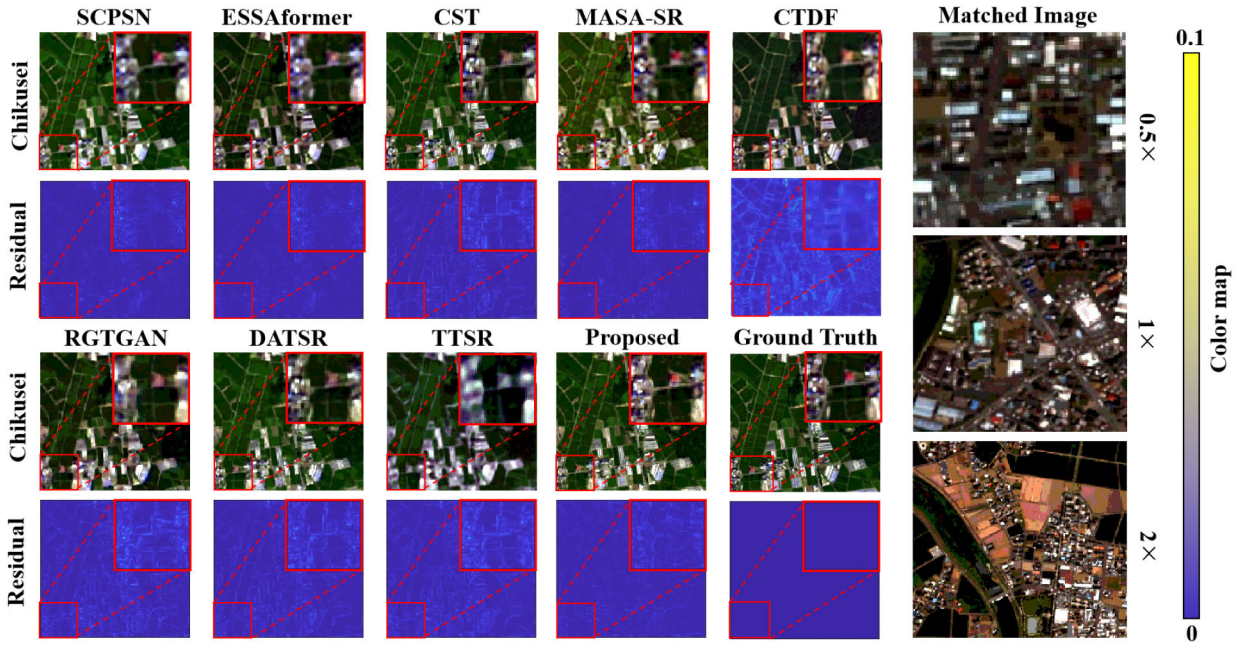


Fig. 6. Visualization results of different methods on Chikusei dataset with GT.

TABLE I
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE HOUSTON, PAVIA, AND CHIKUSEI DATASETS

Dataset	Houston					Pavia					Chikusei				
Method	PSNR	CC	SAM	RMSE	ERGAS	PSNR	CC	SAM	RMSE	ERGAS	PSNR	CC	SAM	RMSE	ERGAS
SCPSN	41.5280	0.9816	2.0555	0.0085	1.7041	37.2223	0.9809	4.2272	0.0142	2.2833	38.5828	0.9601	1.7829	0.0119	3.2369
ESSAformer	36.7253	0.9516	3.3474	0.0147	2.8368	34.5291	0.9696	4.9082	0.0193	3.0084	38.8381	0.9599	1.9037	0.0115	3.3781
CST	39.9451	0.9772	1.9618	0.0102	2.0789	34.1082	0.9658	4.5950	0.0204	3.1756	35.5140	0.9251	2.3101	0.0169	4.3870
MASA-SR	38.3504	0.9642	3.2040	0.0122	2.3749	36.8623	0.9779	4.8016	0.0147	2.4366	40.2761	0.9613	1.8934	0.0098	3.4880
CTDF	38.1201	0.9607	6.0301	0.0126	3.1891	34.1909	0.9702	5.9701	0.0186	3.2906	38.6834	0.9568	4.3867	0.0116	2.5091
RGTGAN	41.0225	0.9788	2.2754	0.0092	1.8181	35.2239	0.9724	4.4911	0.0182	2.7714	34.2872	0.8941	2.9163	0.0195	5.2305
DATSR	43.3583	0.9867	1.8672	0.0068	1.4338	36.1491	0.9767	4.6735	0.0160	2.5838	35.1302	0.9159	2.5905	0.0177	4.6700
TTSR	35.2239	0.9724	4.4911	0.0182	2.7714	34.1538	0.9663	5.8631	0.0201	3.1733	35.0906	0.9299	4.7249	0.0177	3.4300
Proposed	**49.2473**	**0.9950**	**1.0523**	**0.0034**	**0.9134**	**38.3709**	**0.9867**	**3.9669**	**0.0113**	**2.1279**	**43.5335**	**0.9815**	**1.4169**	**0.0067**	**1.4449**

TABLE II
QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON THE XDU-LIYUKOU DATASET

Dataset	XDU-Liyukou				
Method	PSNR	CC	SAM	RMSE	ERGAS
SCPSN	46.9920	0.9928	1.0944	0.0044	0.7510
ESSAformer	44.8048	0.9903	1.3788	0.0044	1.2659
CST	46.9963	0.9913	1.3554	0.0046	0.9341
MASA-SR	46.7426	0.9947	1.4771	0.0048	0.9177
CTDF	40.0672	0.9724	2.2697	0.0070	1.3415
RGTGAN	44.3774	0.9910	2.3168	0.0060	1.1337
DATSR	45.4409	0.9904	1.7112	0.0055	1.0233
TTSR	41.6988	0.9859	2.3763	0.0082	1.5492
Proposed	**48.0052**	**0.9961**	**0.8024**	**0.0041**	**0.6944**

on local spatial alignment and are effective only when the reference image exhibits high regional similarity. Moreover, RGTGAN and DATSR demonstrate degraded performance on the Chikusei dataset, in some cases performing even worse than SISR baselines. TTSR also struggles to outperform SISR methods because of its incompatibility with complex hyperspectral characteristics. In addition, we have conducted the comparison experiments on the real-word dataset (XDU-Liyukou dataset). As shown in Table II, the proposed method achieves the best performance across all five evaluation metrics. As shown in Table III, in the absence of GT, the proposed method achieves the lowest value of D_λ compared with other methods, indicating that MFME-DiffNet maintains superior spectral consistency in real-world scenarios. Our method achieves superior and more stable reconstruction performance across all datasets, demonstrating strong robustness and generalization under challenging unpaired fusion scenarios. In addition to the quantitative improvements, the proposed method exhibits clear qualitative advantages. Compared with others, our method produces reconstructions with finer texture details, more accurate edge structures, and better preservation of small objects. In particular, in regions with complex spectral

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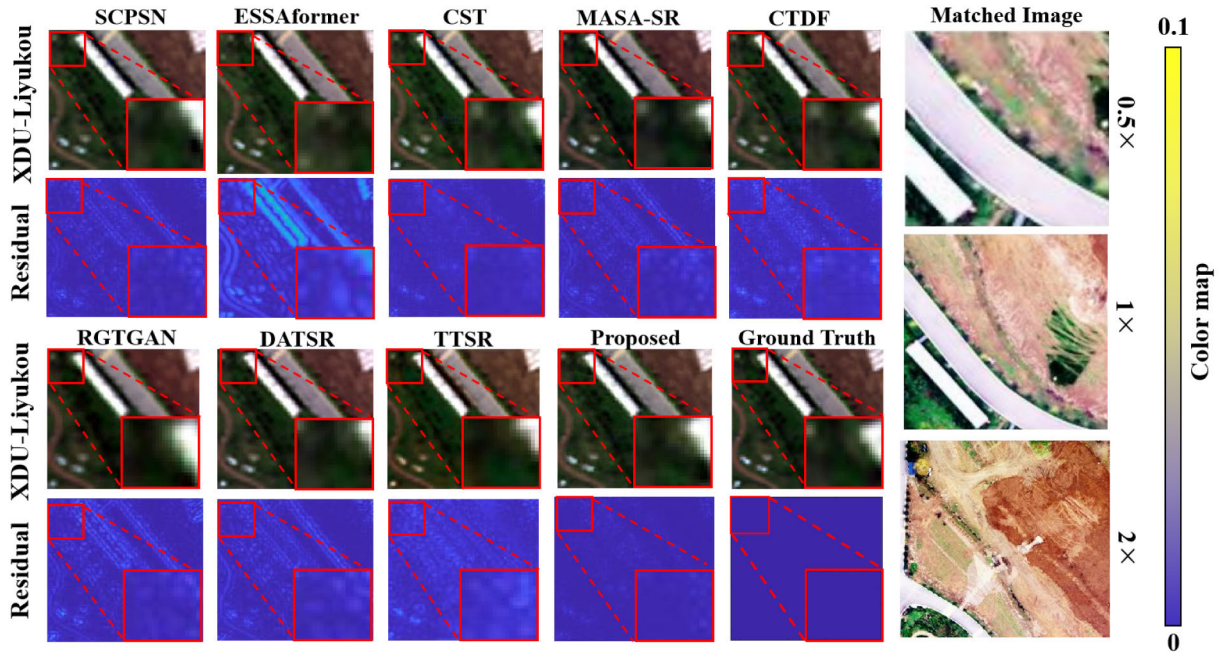


Fig. 7. Visualization results of different methods on XDU-Liyukou dataset with GT.

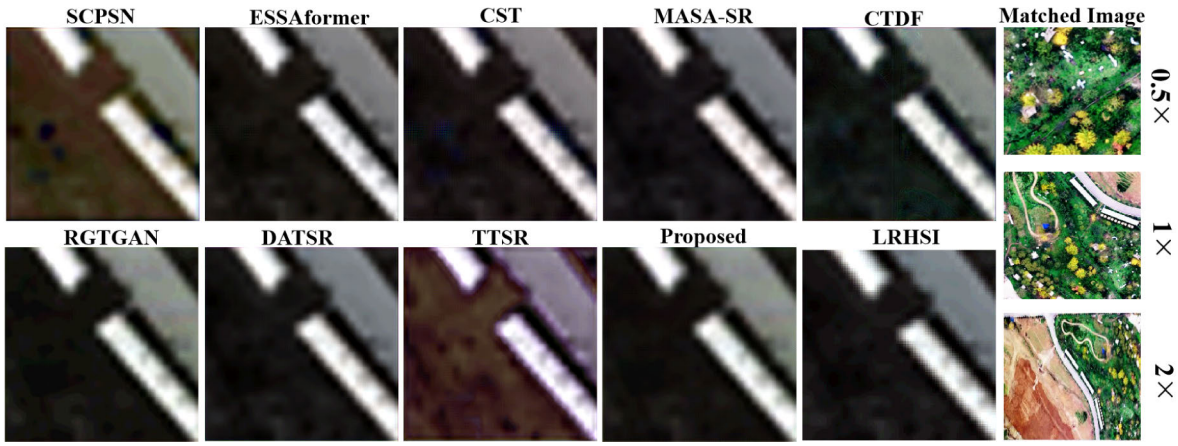


Fig. 8. Visualization results of different methods onthe XDU-Liyukou dataset without GT.

variations or spatial misalignments, our method demonstrates stronger robustness and generates more consistent and realistic hyperspectral results. Moreover, multilevel matching enables improved recovery of semantic structures across scales, yielding higher global fidelity and richer local details. To further assess reconstruction accuracy, we visualize residual maps computed as the mean absolute error across all spectral bands between the reconstructed and reference images. For the Houston and Pavia datasets, representative local regions are enlarged, and residual values are scaled for clarity. As shown in Figs. 4–7, the proposed method consistently exhibits the lowest residuals across all datasets, indicating more faithful spatial and spectral reconstruction. Moreover, Fig. 8 shows that the results of SCPSN and TTSR suffer from severe spectral distortions, whereas the proposed method achieves higher spectral fidelity and clearer spatial details, further demonstrating its robustness and practicality.

D. Ablation Study

1) Effectiveness of GAMM: To systematically evaluate the effectiveness of the proposed GAMM, in which the matching scores are calculated using ISSIM, we conducted an ablation study. One variant is constructed by removing the GAMM module (i.e., w/o GAMM). Another variant is constructed by substituting the ISSIM similarity measure in GAMM with the original SSIM (i.e., w/ SSIM). The comparison results between the proposed method and its variants across three benchmark datasets are presented in Fig. 9. Quantitative results show that our method consistently outperforms the ablated model across key metrics, fully demonstrating the effectiveness of both the GAMM module and the designed ISSIM in generating matched images with features more closely aligned to the target images, thereby significantly improving HSI reconstruction quality. This demonstrates the effectiveness of GAMM.

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TABLE III
EVALUATION RESULTS OF DIFFERENT METHODS ON METRIC D_λ

Method	SCPSN	ESSAformer	CST	MASA-SR	CTDF	RGTGAN	DATSR	TTSR	**Proposed**
D_λ	0.3060	0.2279	0.2620	0.2529	0.2603	0.2445	0.2312	0.2770	**0.2120**

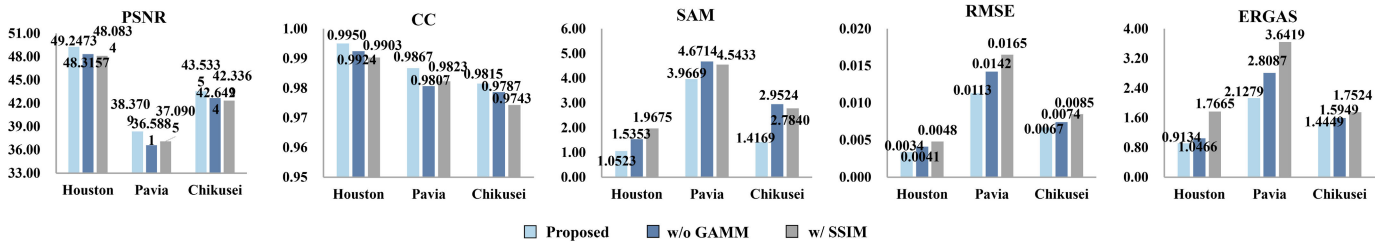


Fig. 9. Performance comparison between the proposed method and its variants (w/o GAMM and w/ SSIM).

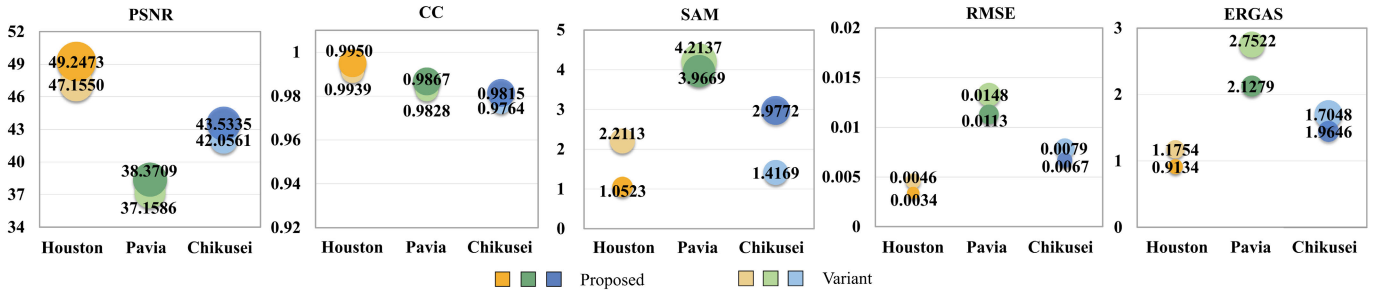


Fig. 10. Impact of iterative matched image update on image fusion performance.

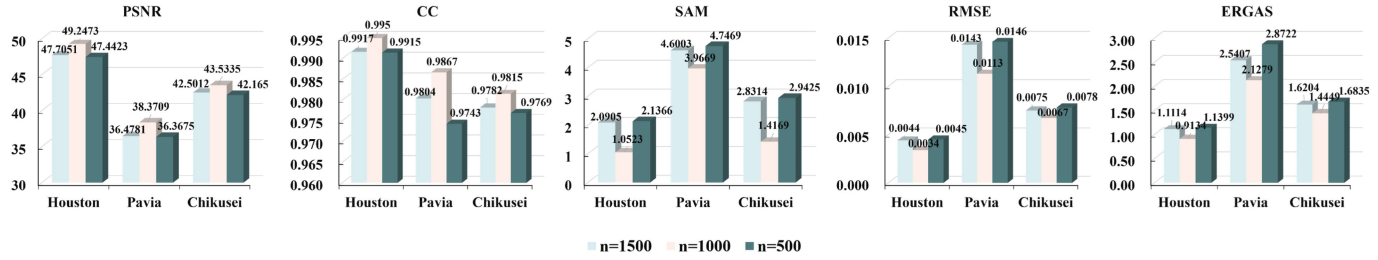


Fig. 11. Experimental results on the impact of dynamic matching start time step.

2) Effectiveness of SWAF: To validate the effectiveness of SWAF, we conducted an ablation study with two variants: one using only single-scale matched images (1×) as guidance (w/ single-scale), and the other without similarity weight (w/o similarity weight). As shown in Table IV, the experimental results demonstrate that our method consistently outperforms the variants across all evaluation metrics, confirming the effectiveness of the proposed SWAF. Compared to single-scale fusion approaches, SWAF uses a progressive fusion strategy to integrate multilevel matched features, achieving more effective feature aggregation and significantly improving the final image fusion quality. Furthermore, the method without the similarity weight exhibits a significant performance decline. This is because the similarity weight ensures the effective fusion of highly similar regions while mitigating the adverse effects caused by dissimilar areas.

3) Effectiveness of Matched Image Update Strategy: To validate the effectiveness of the iterative matching-fusion strategy, we designed a comparative variant that adopts single matching, and conducted systematic experiments on multiple benchmark datasets, with results shown in Fig. 10. The experimental results demonstrate that, compared to single matching methods, the proposed progressive matching-fusion strategy achieves significant improvements across multiple evaluation metrics, delivering finer texture detail recovery and more accurate structural reconstruction. These results confirm that by progressively optimizing the matched images during fusion, our method effectively mitigates the negative impact of initial matching errors and further enhances the overall quality and robustness of image fusion.

4) Effect of Dynamic Matching Start Time Step: To investigate the impact of the start time step n of dynamic matching on HSI reconstruction, we conducted ablation experiments with different start points. Specifically, the total diffusion time step is set to 2000, and dynamic matching is initiated at steps 1500, 1000, and 500, respectively. For the remaining time steps, the pre-matched images obtained from the LRHSI and HRMSI were used as guidance. As shown in Fig. 11,

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TABLE IV
PERFORMANCE COMPARISON BETWEEN THE PROPOSED METHOD AND ITS VARIANTS (W/ SINGLE-SCALE AND W/O SIMILARITY WEIGHT)

Dataset	Method	PSNR	CC	SAM	RMSE	ERGAS
Houston	**MFME-DiffNet**	**49.2473**	**0.9950**	**1.0523**	**0.0034**	**0.9134**
	w/ Single-Scale	48.2757	0.9924	1.9745	0.0041	1.0506
	w/o Similarity Weight	44.0145	0.9895	1.3005	0.0053	1.4973
Pavia	**MFME-DiffNet**	**38.3709**	**0.9867**	**3.9669**	**0.0113**	**2.1279**
	w/ Single-Scale	37.0469	0.9823	4.3137	0.0134	2.7848
	w/o Similarity Weight	35.2580	0.9775	5.1587	0.0177	3.5165
Chikusei	**MFME-DiffNet**	**43.5335**	0.9815	**1.4169**	**0.0067**	**1.4449**
	w/ Single-Scale	42.3740	0.9776	2.8725	0.0076	1.6442
	w/o Similarity Weight	40.5142	**0.9828**	2.2706	0.0080	2.3002

although starting dynamic matching at step 1500 results in a longer matching period, the initial fusion outputs from SWAF contain a large amount of noise, leading to low matching accuracy and insufficient texture guidance for fusion. Conversely, initiating dynamic matching at step 500 provides a shorter dynamic matching duration, causing most time steps to rely on pre-matched images for guidance, which limits the benefits of dynamic matching and negatively affects overall fusion quality. Through comprehensive comparison, we found that starting dynamic matching at step 1000 strikes a better balance between matching accuracy and fusion performance, significantly improving reconstruction quality. Therefore, the starting time step n for dynamic matching is set to 1000.

V. CONCLUSION

In this article, we propose an MFME-DiffNet, which uses multiscale matched images that are similar to HSI in spatial texture to guide the fusion process, and combines the diffusion model to iteratively refine the matching images. Even under the condition of scene differences between HSI and MSI, the fusion quality can still be effectively improved. The proposed GAMM performs feature alignment using gradient features, producing high-quality guidance images. Meanwhile, the SWAF module adaptively fuses these auxiliary cues based on their texture similarity with the target HSI and suppresses interference from semantically dissimilar inputs. The two modules are jointly optimized in an iterative reverse diffusion framework, allowing the fusion results to continuously refine the matching process. This closed-loop design leads to progressive enhancement of spatial details and overall reconstruction quality. MFME-DiffNet achieved outstanding performance on all four datasets, confirming its effectiveness and superiority. The current design is tailored for unpaired optical images. In future work, we plan to explore a more generalizable fusion framework capable of handling unpaired images from diverse modality combinations.

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